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Introduction

The purpose of this repository is to showcase a time-series forecasting model that predicts store sales of a major supermarket corporation based in South America. This Kaggle "Getting Started" competition runs indefinitely with a rolling leaderboard. My goal is to build a model that more accurately predicts the unit sales for thousands of products sold at various Favorita stores. I am also looking to gain a deeper understanding of the data and the corporation through a full statistical analysis. This will give me the opportunity to practice my machine-learning skills, enter my first competition on Kaggle, and gain real-world experience.

Data

As an individual participating in this competition, I will need to predict sales for the thousands of product families sold at Favorita stores located in Ecuador. The training dataset includes information on the dates, stores, and products sold, whether a product was being promoted, and the sales numbers. Additionally, there are supplementary files that may be useful in building my models. **These are the datasets used:**

▼ 1. train.csv dataset:

Column Name	Description
store_nbr	Identifies the store at which the products are sold
family	Identifies the type of product sold
sales	Provides the total sales for a product family at a particular store on a given date. Note that fractional values are possible since products can be sold in fractional units (e.g., 1.5 kg of cheese)
onpromotion	Indicates the total number of items in a product family that were being promoted at a store on a given date

▼ 2. test.csv dataset:

Column name	Description
id	Unique ID for the row
date	Date of the sales forecast
store_nbr	Identifies the store at which the products are sold

Column name	Description
family	Identifies the type of product sold
onpromotion	Total number of items in a product family that were being promoted at a store on a given date

▼ 3. sample_submission.csv dataset:

Column name	Description
id	Unique ID for the row
sales	Predicted sales for the corresponding id in test.csv

▼ 4. stores.csv dataset:

Column name	Description
store_nbr	Identifies the store
city	City where the store is located
state	State where the store is located
type	Type of store (A, B, or C)
cluster	Grouping of similar stores

▼ 5. oil.csv dataset:

Column name	Description
date	Date of the oil price
dcoilwtico	Daily oil prices (West Texas Intermediate)

▼ 6. holidays_events.csv dataset:

Column name	Description
date	Date of the holiday/event
type	Type of holiday/event (Holiday, Additional, Bridge, or Transfer)
locale	Locale of the holiday/event (National, Regional, or Local)
locale_name	Name of the holiday/event locale
description	Description of the holiday/event
transferred	Indicates whether a holiday was officially transferred to another date by the government

Citation Alexis Cook, DanB, inversion, Ryan Holbrook. (2021). Store Sales - Time Series Forecasting. Kaggle.